Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/11/2020
PI Name:	Palaima, Elizabeth Ph.D.		
Project Title:	Room Temperature Stable Basic Metabolic Panel		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	epalaima@ativamed.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	617-818-6924
Organization Name:	Ativa Medical Corporation		
PI Address 1:	1000 Westgate Dr.		
PI Address 2:	Suite 100		
PI Web Page:			
City:	Saint Paul	**State:**	MN
Zip Code:	55114	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
Start Date:	05/01/2019	**End Date:**	03/31/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2020 as project ended early, per TRISH (Ed., 9/10/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-IND0006		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Industry Project Ativa's mission is to bring common laboratory tests to the point of care. This proposal is for Ativa's second module: a cost effective and room temperature storage basic metabolic panel (BMP) test using electrochemistry. The project proposed incorporates temperature stabilized enzymes into our electrochemistry development, specifically for a more robust sensor in order to avoid refrigeration. This BMP will be incorporated into Ativa's pre-market flow cytometry system. The Ativa system will be able produce a complete blood count (CBC) and BMP test with clinical sensitivity and specificity comparable to that of a conventional laboratory. This validation project will produce a creatinine sensor with a shelf life of one month, supporting validation of the initial temperature stable innovation and its subsequent electrosensor prototype execution. The final commercial product has a target shelf life of over two years and will require		

	no cold chain refrigeration. The Ativa system is intuitive to use and designed for personnel who have little or no experience running blood tests. The device will accept blood from finger stick, heel stick, or venous draw into a disposable microfluidic card, eliminating the need for a trained phlebotomist. A touch screen interface will walk the user through a step by step process with visual assistance demonstrating the sample loading procedure and card handling. Once the test is completed, there will be numerically and graphically displayed ranges which show whether the test result falls within the normal range.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work supports the promise of a small point of care, fingerstick capable, combined BMP CBC test device for non medical users.
Task Progress:	Progress was made toward producing a small point of care, fingerstick capable, combined basic metabolic panel (BMP) complete blood count (CBC) test device for non-medical users. However, the project was closed by the Translational Research Institute for Space Health due to multiple operational factors that affected the company's ability to deliver on their aims.
Bibliography Type:	Description: (Last Updated:)